Introduction to Linux

What is Linux?

Linux

Linux is, in simplest terms, an **operating system**, assembled under the model of free and open source software development and distribution.

Operating System

Operating systems are computer programs. An operating system is the first piece of software that the computer executes when you turn the machine on. The **operating system** loads itself into memory and begins managing the resources available on the computer. It then provides those resources to other applications that the user wants to execute.

To say that **Linux** is an **operating system** means that it's meant to be used as an alternative to other operating systems, **Windows**, **Mac OS**, **MS-DOS**, **Solaris** and others. **Linux** is not a program like a word processor and is not a set of programs like an office suite. Linux is an interface between computer/server hardware, and the programs which run on it.

Why Linux?

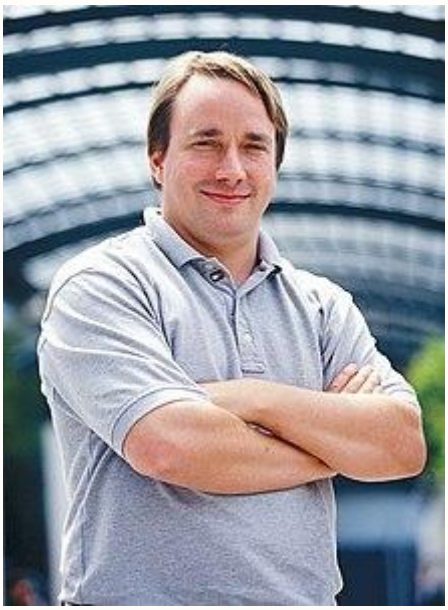
Linux has proven to be a tremendously stable and versatile operating system, particularly as a network server. Its **down-time** is almost negligible. We can run Linux servers continuously without reboot for many years. There have been cases when Linux servers have been running for more than 3-4 years without re-booting and then only taken down for a brief period for routine maintenance.

History of Linux

Linux began in 1991 as a personal project by Finnish student Linus Torvalds to create a new free operating system kernel. The resulting Linux kernel has been marked by constant growth throughout its history.

In 1991, while studying computer science at University of Helsinki, Linus Torvalds began a project that later became the Linux kernel. He wrote the program specifically for the hardware he was using and independent of an

operating system because he wanted to use the functions of his new PC with an 80386 processor. Development was done on MINIX using the GNU C Compiler.



Linus Torvalds in 2002

Linux is an **open-source operating system** that powers millions of devices around the world, from servers and desktop computers to smartphones, routers, and even embedded systems. It's based on the **Linux kernel**, which was initially created by Linus Torvalds in 1991. Since then, Linux has grown into a family of operating systems, collectively referred to as **Linux distributions** , each tailored for different use cases.

Key Characteristics of Linux:

1. **Open Source**:
    - **Freedom and Customization**: Linux is free and open-source software, meaning anyone can access the source code, modify it, and distribute their own versions. This allows users and organizations to tailor Linux to their specific needs.
2. **The Linux Kernel**:
    - The **kernel** is the core of the operating system. It manages hardware resources (CPU, memory, and devices), provides process management, and facilitates communication between software and hardware.
3. **Linux Distributions**:

- A **distribution** is a package of the Linux kernel along with various software tools, libraries, and user interfaces. Popular.

Linux is the most popular operating system which provides several powerful administrative tools and utilities that help us to manage the systems effectively. With the Linux command line feature we can access any system remotely and securely.

Linux OS has many distributor. Some are the following.

Ubuntu:
Currently the most popular distributor. The most user-friendly version for Linux newbies.

Fedora:
The next most popular distributor is Red Hat's free distributor that includes a powerful desktop GUI based on Gnome 3 but there are also various spins of it that use different desktops - such as KDE, LXDE, XFCE. It is often amongst the first to include new technology and is strong on security. It has strong command line feature and is more reliable and ideally suited to the slightly more adventurous user.

Linux Mint:
It is another user friendly version of Linux, based on the Ubuntu core or can described as "an improved Ubuntu".

CentOS:
It is completely based on Red Hat Enterprise Linux. It is mostly used for servers.
Red Hat:
It is the best server *distribution*. But it is not free; you have to pay for it. If an organization doesn't mind spending dollars on purchasing the red-hat support, this is no 1 recommendation to any organization who runs mission critical applications.

Debian:
Debian is known as one of the most well-tested and bug-free *distributor*. It serves as the foundation for Ubuntu, most view Debian as a distro best-suited for those experienced with Linux. Debian has a relatively slow release cycle, with stable ones coming out every one to three years.

4. **Command Line Interface (CLI)**:
   - While many Linux distributions provide graphical user interfaces (GUIs), Linux is highly powerful when used through its command-line interface (CLI). The CLI allows users to perform a wide range of tasks such as file management, system administration, and software development with precision and flexibility.

5. **Multi-User and Multi-Tasking**:
    - Linux is designed to be a **multi-user** operating system, meaning multiple users can access and use the system simultaneously. It also supports **multi-tasking**, enabling several programs to run at the same time.
6. **Security**:
    - Linux is known for being secure. It has a strong security model, frequent updates, and the open-source community constantly audits and patches vulnerabilities. Linux also provides advanced user permissions and access controls, making it difficult for unauthorized users to exploit the system.
7. **Flexibility and Performance**:
    - Linux can be used on a wide range of devices, from supercomputers to tiny embedded systems. Its performance is highly optimized, making it a preferred choice for tasks requiring high computing power and low latency (e.g., servers, scientific computing, and real-time systems).

Common Uses of Linux:

1. **Servers**: Linux powers the majority of web servers, database servers, and cloud infrastructure due to its reliability, security, and scalability.
2. **Desktop Computers**: Many users choose Linux desktops (such as **Ubuntu** or **Mint**) for personal computing. It is free, secure, and lightweight, making it ideal for older hardware or privacy-conscious users.
3. **Cloud and Virtualization**: Linux is widely used in cloud platforms and for virtualization technologies, like **Docker** and **Kubernetes**, which allow for efficient and scalable application deployment.
4. **Development**: Developers prefer Linux for its rich set of programming tools, package managers, and open-source libraries.
5. **Embedded Systems**: Linux powers devices like routers, smart TVs, automotive systems, and industrial control systems due to its lightweight nature and ability to run on a wide variety of hardware.
6. **Mobile**: **Android**, the world's most popular mobile operating system, is based on the Linux kernel.

Summary:

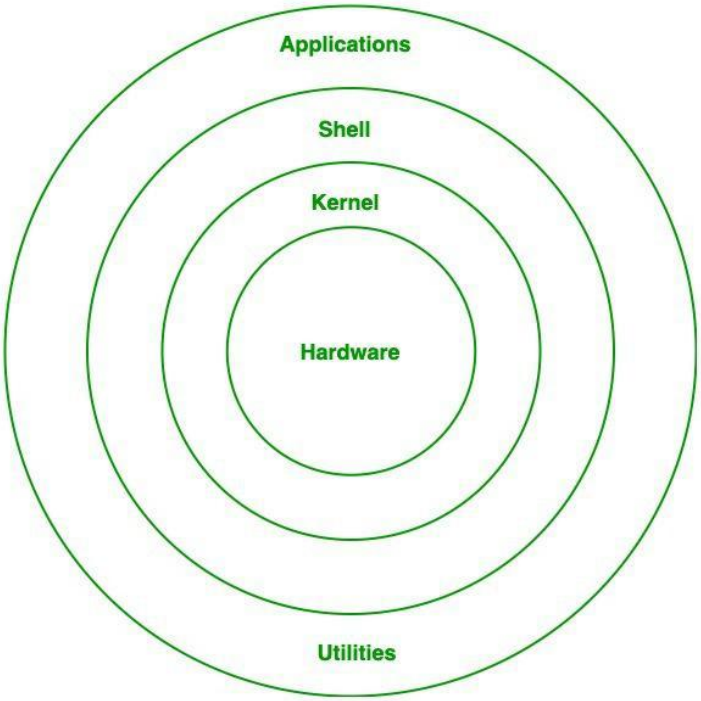
Linux is a powerful, secure, and highly customizable operating system that serves a broad range of applications from personal computing to enterprise-level systems and devices. Its open-source nature ensures constant innovation, making it a foundational technology across many industries.

Architecture of linux operating system

Components of Linux:

Like any operating system, Linux consists of software, computer programs, documentation, and hardware.

The main components of Linux operating system are: Application, Shell, Kernel, Hardware, Utilities



Linux operating system architecture

1. Hardware: The lowest level of the Linux architecture is the hardware layer. This layer comprises the physical components of a computer, such as the hard drive, RAM, motherboard, CPU, network interfaces, and

peripherals. These components are the tangible pieces of your system on which the rest of the architecture is built.

2. Kernel: Directly interfacing with the hardware layer is the kernel, the heart of the Linux operating system. As the core part of the OS, the kernel is responsible for low-level tasks such as disk management, task scheduling, memory management, and controlling peripherals.

3. Shell: One layer up from the kernel is the shell. In simplest terms, the shell is a user interface that allows users to interact with the kernel. In Linux, most interactions with the shell occur in a command-line interface (CLI), where users type commands interpreted by the shell.

There are several different shells available in Linux, each with its unique features and syntax, such as

1. **The Bourne Shell (sh)**
2. **Bash (Bourne Again Shell)**
3. **C Shell (csh)**
4. **TC Shell (tcsh)**
5. **Korn Shell (ksh)**

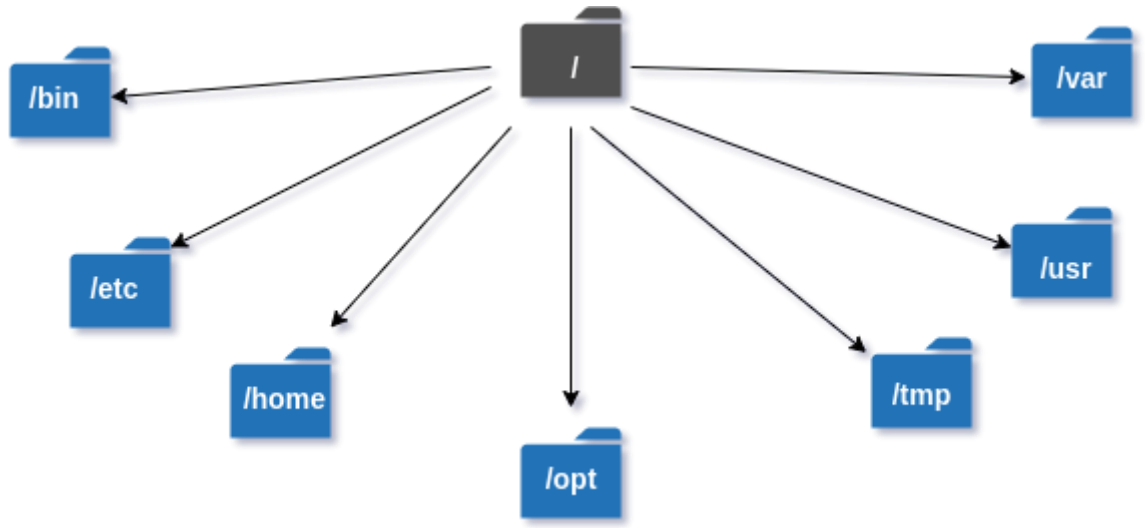
4. Applications: The topmost layer of the Linux architecture consists of applications. These are the software programs that you, as the user, interact with directly. They range from system applications like file managers, text editors, and network managers, to user applications like browsers.
While applications communicate with the hardware through the kernel, they interact with the user through the shell. For instance, when you run a command to open a file in a text editor, the shell interprets your command, the kernel fetches the file from the hardware, and the text editor (application) displays it.

In Linux/Unix operating system everything is a file even directories are files, files are files, and devices like mouse, keyboard, printer, etc are also files. Here we are going to see the Directory Structure in Linux.

Types of files in the Linux system.

1. **General Files** – It is also called ordinary files. It may be an image, video, program, or simple text file. These types of files can be in ASCII or Binary format. It is the most commonly used file in the Linux system.
2. **Directory Files** – These types of files are a warehouse for other file types. It may be a directory file within a directory (subdirectory).
3. **Device Files –** In a Windows-like operating system, devices like CD-ROM, and hard drives are represented as drive letters like F: G: H whereas in the Linux system devices are represented as files. As for example, /dev/sda1, /dev/sda2, and so on.

We know that in a Windows-like operating system, files are stored in different folders on different data drives like C: D: E: whereas in the Linux/Unix operating system files are stored in a tree-like structure starting with the root directory as shown in the below diagram.

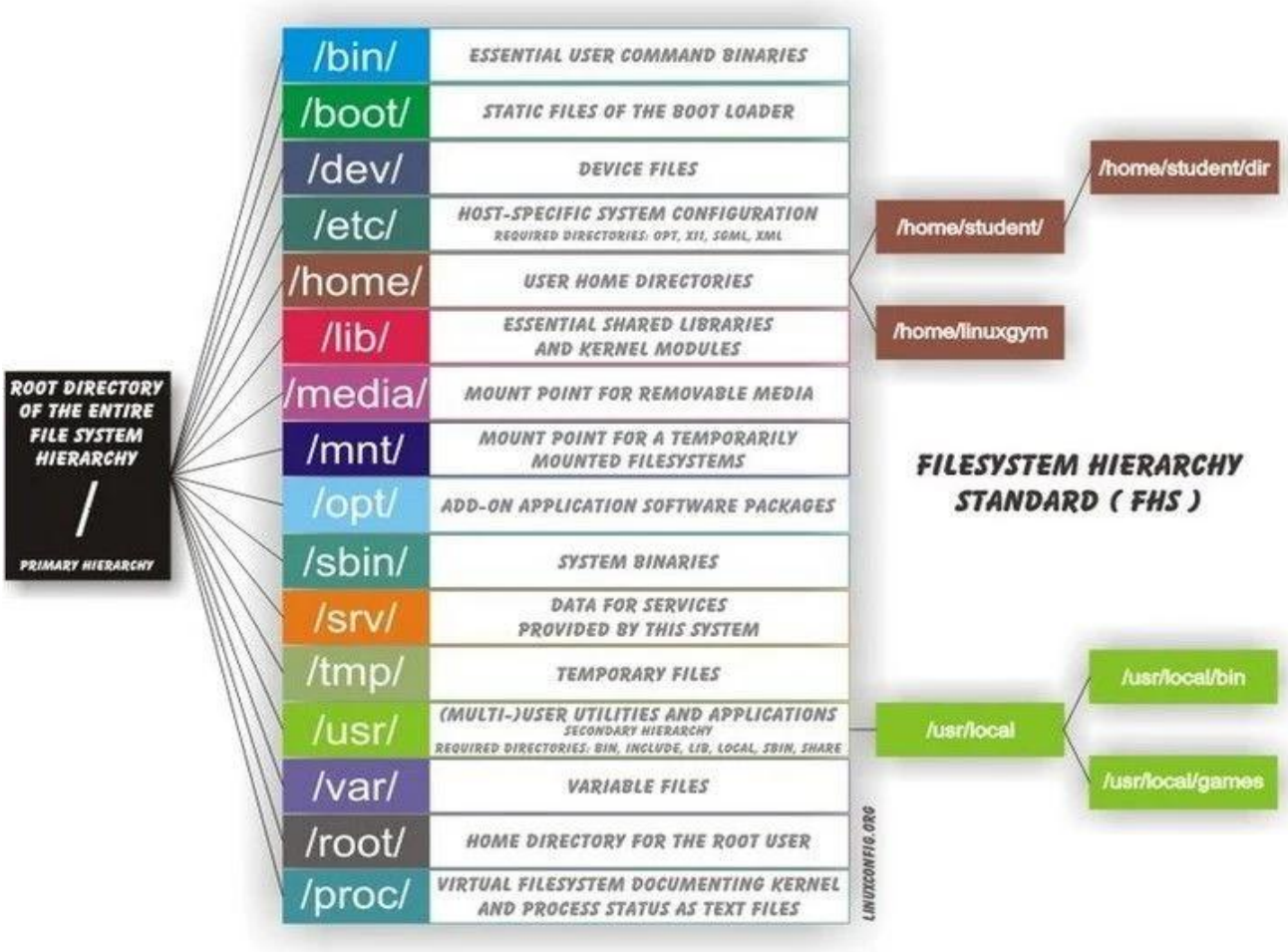


data storage in Linux/Unix operating systems

The Linux/Unix file system hierarchy base begins at the root and everything starts with the root directory.

Linux File system/File Hierarchy Standard (FHS)

The Linux File Hierarchy Structure or the File system Hierarchy Standard (FHS) defines the directory structure and directory contents in Unix-like operating systems. It is maintained by the Linux Foundation.



These are the common top-level directories associated with the root directory:

Directories	Description
/bin	binary or executable programs.
/etc	system configuration files.
/home	home directory. It is the default current directory.
/opt	optional or third-party software.
/tmp	temporary space, typically cleared on reboot.
/usr	User related programs.
/var	log files.

Some other directories in the Linux system:

Directories	Description
/boot	It contains all the boot-related information files and folders such as conf, grub, etc.
/dev	It is the location of the device files such as dev/sda1, dev/sda2, etc.

Directories	Description
/lib	It contains kernel modules and a shared library.
/lost+found	It is used to find recovered bits of corrupted files.
/media	It contains subdirectories where removal media devices are inserted.
/mnt	It contains temporary mount directories for mounting the file system.
/proc	It is a virtual and pseudo-file system to contains info about the running processes with a specific process ID or PID.
/run	It stores volatile runtime data.
/sbin	binary executable programs for an administrator.
/srv	It contains server-specific and server-related files.
/sys	It is a virtual file system for modern Linux distributions to store and allows modification of the devices connected to the system.

1. / (Root):

Primary hierarchy root and root directory of the entire file system hierarchy.

- Every single file and directory start from the root directory.
- The only root user has the right to write under this directory.
- /root is the root user's home directory, which is not the same as /

2. /bin :
Essential command binaries that need to be available in single-user mode; for all users, e.g., cat, ls, cp.

- Contains binary executables.
- Common linux commands you need to use in single-user modes are located under this directory.
- Commands used by all the users of the system are located here e.g. ps, ls, ping, grep, cp

3. /boot :
Boot loader files, e.g., kernels, initrd.

- Kernel initrd, vmlinux, grub files are located under /boot

4. /dev :
Essential device files, e.g., /dev/null.

- These include terminal devices, usb, or any device attached to the system.

5. /etc :
Host-specific system-wide configuration files.

- Contains configuration files required by all programs.
- This also contains startup and shutdown shell scripts used to start/stop individual programs.

6. /home :
Users' home directories, containing saved files, personal settings, etc.

- Home directories for all users to store their personal files.

7. /lib:
Libraries essential for the binaries in /bin/ and /sbin/.

- Library filenames are either ld* or lib*.so.*

8. /media:
Mount points for removable media such as CD-ROMs

- Temporary mount directory for removable devices.
- Examples, /media/cdrom for CD-ROM; /media/floppy for floppy drives

9. /mnt :
Temporarily mounted filesystems.

- Temporary mount directory where sysadmins can mount filesystems.

10. /opt :
Optional application software packages.

- Contains add-on applications from individual vendors.
- Add-on applications should be installed under either /opt/ or /opt/ sub-directory.

11. /tmp :
Temporary files. Often not preserved between system reboots and may be severely size restricted.

- Directory that contains temporary files created by system and users.
- Files under this directory are deleted when the system is rebooted.

12. /usr :
Secondary hierarchy for read-only user data; contains the majority of (multi-)user utilities and applications.

13. /proc:
Virtual filesystem providing process and kernel information as files. In Linux, it corresponds to a procs mount. Generally, automatically generated and populated by the system, on the fly.

- Contains information about system process.

Difference between Unix and Linux:

- **Unix** is a commercial, proprietary operating system originally designed in the 1970s, used mainly for enterprise servers and workstations.
- **Linux** is an open-source, Unix-like operating system kernel that has evolved into a family of free operating systems (GNU/Linux), used in everything from personal computers and servers to embedded systems and mobile devices.

- **Unix**: Unix systems are typically **proprietary** and require users to purchase licenses for enterprise use (especially for commercial Unix variants).

- **Linux**: **Free to use**, though certain enterprise Linux distributions (like **Red Hat Enterprise Linux** or **SUSE Linux Enterprise**) offer paid support options.

Absolute Path Vs Relative Path In Linux

Absolute Path Vs Relative Path In Linux:

Absolute Path: An *absolute path* is defined as specifying the location of a file or directory from the root directory(/). In other words we can say *absolute path* is a complete path from start of actual filesystem from / directory.

example:

/home/user/Document/srv.txt

Relative Path: *Relative path* is defined as path related to the present working directory(pwd). Suppose I am located in /home/user1 and I want to change directory to /home/user1/Documents. I can use *relative path* concept to change directory to Documents.

example:

here are two examples for changing directory, 1st by using relative path, 2nd by using absolute path.

$ pwd
/home/user1
$cd Documents/ (using relative path)
$pwd
/home/user1/Documents

Proprietary refers to something that is owned by an individual or a company and is subject to specific legal rights or restrictions. In the context of technology, software, hardware, or products that are proprietary are typically closed-source and controlled exclusively by the owner or creator. Users or consumers must adhere to certain conditions or licenses to use, modify, or distribute them.

Key Features of Proprietary Software or Technology:

- **Ownership**:
    - Proprietary software or technology is owned by a specific company, developer, or individual. The source code or inner workings are not publicly available.
- **Closed Source**:
    - Proprietary software is typically **closed-source**, meaning the source code is not accessible to the public. Users cannot modify or inspect the underlying code.
- **Licensing**:

- Users of proprietary software must agree to the **license terms** set by the owner, which typically restrict how the software can be used, modified, or redistributed. A common example is **end-user license agreements (EULAs)**.
- License fees may apply, and the software might come with restrictions on the number of installations or the types of devices it can be used on.

- **Limited Modification**:

  - Users are generally **not allowed** to modify proprietary software. Only the owner or authorized developers can make changes or updates.

- **Support**:

  - Proprietary software often comes with **vendor-provided support** or updates, usually at a cost or as part of a licensing agreement.

- **Distribution Control**:

  - The owner of the proprietary software controls how it is distributed and can limit its availability, restricting users from sharing the software with others.

Examples of Proprietary Software:

- **Microsoft Windows**: A proprietary operating system developed by Microsoft. Users need to purchase a license to use it, and the source code is not available for modification.
- **Adobe Photoshop**: A proprietary image-editing software that requires users to purchase or subscribe to it. The source code is closed, and users can't modify or redistribute the software.
- **MacOS**: Apple's proprietary operating system, which is only licensed for use on Apple hardware.

Most commonly Used Text Editors in Linux

For Linux users, text editors are essential tools that play a crucial role in various tasks, from coding and writing to system administration and configuration management. Linux offers a wide range of text editors, catering to different preferences and use cases. In this article, we will delve into the world of Linux text editors, exploring their features, strengths, and popular choices among the community.

Mostly Used Text Editors in Linux

- Vi Text Editor
- Vim Editor
- Nano Editor

The Linux shell is the bridge that connects the user with the operating system. At the core of the Linux operating system lies the kernel. This kernel is responsible for managing the resources of the computer, like the memory, CPU, etc. When the user wants to make the kernel work, it has to connect to it through a shell. **Linux shell**

programming is the term associated with shell scripting, which uses simple English to give instructions to the computer. You can even instruct the operating system to do repetitive tasks through **Linux shell programming**.

Note:

*System hard disk is named as cygdrive

*All the drives of my windows OS will be present under/cygdrive

*All directories (folders) are accessed using '/'(delimiter) separation

*In Linux 'Tab'key play an important roll

*tab button on keyboard is used to automatically fill the paths and names

*bash:bourne again shell

*pwd:present working directory

cd: change directory

cd /:root directory

cd ~:Home directory

cd .. goes to previous directory

cd ../../../ goes to previous directories(3 steps previous)

cd path1/path2/path3/: Absolute Path

cd 'dir name' :if directory name is 'my name'

cd "dir name" :if directory name is "my name"

cd dir\ name :if directory name is my\ name

clear:it clear the terminal screen

ls:list the directory and files

ls /:list of root directory contents

ls ~:list of home directory contents

ls dir_name: list of "dir_name" contents

ls .. list the contents of previous directory

ls -l:long listing format

ls -a: list the hidden file and directories (list all files and directories)

ls -la: long list with hidden directories

ls -sl list the contents according to size

ls -t : sort by time, newest first;

ls -r : reverse order while sorting

ls -ltr:long list with time and reverse sorting

ls -d */:list of only directories

man <command>:manual of given command

q: to exit from manual

File management

---

command used to create files are

*touch

*cat

touch <filename1.ext> <filename2.ext>

touch filename{1..10}.txt

touch directory/filename --> this command create a file in a directory directly

this command creates the file

cat: its echoes what you typed

*cat: concatenate files and print on the standard output

cat file_name:display the contents of the file

cat dir1/filename

cat -b filename : add the line numbers to non-blank lines

cat -n filename : add the line numbers to all lines

cat > filename : '>' this redirects and open the file for edit

cat >> filename : '>>' this appends and edit without overwriting the previous data of file

cat file1 > file2 this command redirect the contents file1 to file2

cat dir1/filename display the content of the filename.

To create different type extention files with different names

---

```
for i in $(seq 1 5)
do
touch file$i.txt File$i.v newfile$i.py Sample$i.c Program$i
done
```

Make Directories

---

mkdir <directory_name>: make directory

mkdir <name1> <name2> <name3>

mkdir <name1>/<name2>

mkdir dir{1..10}

mkdir -p <name1>/{name2,name3,name4}

mkdir -p <name1>/<name2>/<name3>/<name4>

mkdir -pv <name1>/<name2>/<name3>/<name4>

mkdir -p its creates the directories with parent directory with sub directory

mkdir -pv print a message for each created directory

ls -R list out the directory structure

rm & rmdir commands

---

rmdir <dir_name>

rmdir -p <name1>/<name2>/<name3>/<name4>

rmdir -pv <name1>/<name2>/<name3>/<name4>

rm -rv <name1> rv--> means recrussive verbose

copying directories

****

cp :{souce file/dir to destination file/dir}

cp sum1 sum2; [sum1 is copied to sum2]

cp filename{1..10}.txt directory

cp -r dir1 new_dir {it creates a new_dir and copies the content of dir1}

cp -r dir1 dir3 { if dir3 already exists,it will copy the dir1 to dir3}{

cp -i without overwrite

cp ../test1.txt ../test2.txt . from previous directory it takes the files and copy to current directory [dot represent current directory]

Move Directories/files

mv source dir1 destination dir2 [dir1 moved to dir2]

mv dir1 ./dir11/dir12/dir13/dir14 {note '.' represent current directory}in this above command dir1 moved to sub directory dir14

mv -i file3.txt dir1

how to rename directories

mv junk junk1 {here the junk directory renamed as junk1} so 'mv' can be used for rename also

mv example /cygdrive/c/Users/VLSIGURU/Desktop/ this command will move directory 'example' to desktop by the path given my us.

mv filename{1..10}.txt directory

Note: 1.mv command we can move single source_file to single destination_file

2.we can move multiple files to single directory (but the directory should exist)

3.we can move single directory to single directory(its not compulsory that destination directory should exist)

4.we can move multiple directories to single directories(but destination directory should exist)

Miscellaneous Commands

---

1. ls : list all the files and directories.
2. pwd : present working directory.
3. head: display the top 10 lines of the file.
4. Tail : Display the bottom 10 lines of the file.
5. Tac : Display the lines in the reverse order.
6. More : Similar to cat command. We can display large content.
7. Less : we can display small content.
8. Id : Display the ID of the user (or) group.
9. Ping : check the network connectivity, status of network/server.
10. Hostname : Display the owner name.
11. Ps : used to list the currently running processes and their PIDs along with some other information depends on different options.
12. Kill : It is used for manually terminating the processes.
13. who : lets you display the users currently logged in to your UNIX or Linux operating system.
14. Whoami : It displays the username of the current user when this command is invoked.
15. Date :displays current date and time
16. Cal : display current year calender